

Devesh Shah

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ML Scientist with 5+ years researching, building and deploying cutting-edge models at the intersection of AIxScience.

Experience

Matterworks

Oct 2023 – Present

Senior Machine Learning Scientist (Remote, NYC)

Apr 2024 – Present

- Developed the core architecture and self-supervised pretraining regime for the Large Spectral Model (LSM), a multimodal foundation model for mass spectrometry ingesting MS1/MS2 spectra and text-based metadata across 1M+ samples for predictive biology tasks.
- Scaled training across multiple GPUs using model parallelism and memory layout optimization to maximize GPU utilization; adapted the architecture with cross-attention and Perceiver-style compression to process 100M-point cloud inputs from raw MS1 spectra.
- Led end-to-end research and development of a transformer-based models for MS2 retrieval of small molecules and generative ID of lipids across 100K+ analytes, achieving SOTA on the MassSpecGym benchmark [[arXiv](#)].
- Designed a joint curriculum and contrastive learning strategy, selecting difficult isomer examples and mining hard negatives from a MoCo-style memory bank, to disentangle representations of structurally similar molecules.
- Led cross-functional collaboration spanning software, ML engineering, and chemistry teams, taking projects from initial brainstorming through research, software integration, and customer deployment.
- Joined as the 3rd ML researcher post-seed and helped scale the research team from 3 to 8 through Series A; designed the hiring process and reported directly to the CTO following the Head of AI's departure.

Machine Learning Scientist (Onsite, Boston)

Oct 2023 – Apr 2024

- Engineered a physics-grounded supervised model for absolute analyte quantification (from wet lab data collection through customer deployment), beating industry-standard by 20% over the company's full metabolite panel.
- Spearheaded company-wide QC standards and automated filtering algorithms that removed 30% of corrupted samples and improved molecular generalization by 15%.

Ford Motor Company Research & Advanced Engineering, Detroit, MI

Jan 2022 – June 2023

Machine Learning Research Scientist – AI4Science, Core AI/ML/QC Research

- Architected a closed-loop generative-AI platform for materials discovery using GANs, latent space optimization and process-structure deep learning models. Leveraged differential augmentation strategies in training with limited data and complex features from micro-structure images [[NeurIPS Workshop Spotlight Paper](#)].
- Built a synthetic data pipeline using U-Net/GAN based domain transfer to improve accuracy of object detection models by 10% for autonomous driving at nighttime with lens flare.
- Coordinated joint ML/Chemistry project with university partners to research novel battery materials using generative geometric deep learning methods and quantum mechanics (DFT).

Van Lehn Laboratory, University of Wisconsin

May 2021 – Sept 2021

Graduate Research Assistant, Geometric Deep Learning for Molecular Dynamics Simulations

ExxonMobil Research & Engineering, Houston, TX

Aug 2018 – June 2020

Process Modeling & Simulation Engineer, Kinetic & Thermal Analysis for Refinery Optimization

Independent Research & Projects

- Interaction Effects in Physics-Informed Neural Networks for Navier-Stokes Vortex Shedding (Aug 2026) [[arXiv](#)]
- A Systematic Evaluation of OOD Detection Methods For Plant Pathology Classification (May 2026) [[arXiv](#)]
- Analyzing Localization Abilities of Vision Transformers in Deep Learning (Dec 2021) [[paper](#)]

Education

University of Wisconsin, Madison, WI

Dec 2021

M.S. Computer Science, Specialization in Machine Learning; GPA: 3.9/4.0

University of Michigan, Ann Arbor, MI

May 2018

B.S.E. Chemical Engineering, Minor in Computer Science; GPA: 3.9/4.0

Key Skill Sets

- Deep Learning: Transformers, GNN, VAE, Contrastive & Active learning, Self-Supervised Pretraining
- Frameworks: PyTorch, AWS, Autoresearch & Agentic Loops, Multi-GPU training (DDP, Model Parallelism)
- Leadership: Technical Mentorship, Team Building & Hiring, Cross-team Communication, Roadmap Ownership

Selected Publications

- Asher, Gabriel*; **Shah, Devesh***; et al. (2025). “LSM-MS2: A Foundation Model Bridging Spectral Identification and Biological Interpretation.” arXiv preprint arXiv:2510.26715 (2025). [[arXiv](#)]
- Shah, Devesh**; Suresh, Anirudh; et al. (2022). “A Survey on Evaluation Metrics for Synthetic Material Micro-Structure Images from Generative Models.” AI for Accelerated Materials Design NeurIPS 2022 Workshop. [[NeurIPS Workshop Spotlight Paper](#)]
- Suresh, Anirudh; **Shah, Devesh**; et al. (2023). “Multi-Objective Evolutionary Design of Microstructures using Diffusion Autoencoders.” AI for Accelerated Materials Design NeurIPS 2023 Workshop. [[NeurIPS Workshop Paper](#)]
- Shah, Devesh**. (2026). “Complementary, Not Cumulative: Interaction Effects in Physics-Informed Neural Networks for Navier-Stokes Vortex Shedding.” arXiv preprint arXiv:2608.19632 (2026). [[arXiv](#)]
- Shah, Devesh**. (2026). “Beyond Toy Benchmarks: A Systematic Evaluation of OOD Detection Methods For Plant Pathology Classification.” arXiv preprint arXiv:2605.08618 (2026). [[arXiv](#)]
- Liu, Zhenyi; **Shah, Devesh**; et al. (2023). “Using Simulation to Quantify the Performance of Automotive Perception Systems.” arXiv preprint arXiv:2303.00983 (2023). [[arXiv](#)]
- Liu, Zhenyi; **Shah, Devesh**; et al. (2025). “ISETHDR: A Physics-Based Synthetic Radiance Dataset for High Dynamic Range Driving Scenes.” IEEE Sensors Journal, vol. 25, no. 9, pp. 15261–15269 (2025). [[IEEE Xplore](#)]
- Ferro, Luke S.; et al. (2024). “A Scalable Approach to Absolute Quantitation in Metabolomics.” bioRxiv 2024.09.09.609906 (2024). [[bioRxiv](#)]